

Title: Effect of depth of NNGP on the accuracy and MSE of the models

< Maximum accuracy and minimum MSE for each training set size is highlighted with green color. >

Accuracy

σ^2 of w and b = 3.0 and 0.8												σ^2 of w and b = 4.0 and 0.84												σ^2 of w and b = 5.0 and 0.9											
Training set size												Training set size												Training set size											
depth	200	400	600	800	1000	1200	1400	1600	1800	2000	2200	depth	200	400	600	800	1000	1200	1400	1600	1800	2000	2200	depth	200	400	600	800	1000	1200	1400	1600	1800	2000	2200
13	0.8178											13												13											
12	0.818	0.8843	0.9041	0.9137								12												12											
11	0.8184	0.8852	0.9044	0.9144								11												11											
10	0.8181	0.8853	0.9044	0.9154	0.9206	0.9275	0.9313					10	0.8066	0.8763	0.8965	0.9051	0.9125	0.918	0.922	0.9239	0.929	0.9335	0.9356	10											
9	0.8181	0.8859	0.9051	0.9151	0.9213	0.928	0.9322					9	0.8102	0.8778	0.8979	0.9077	0.9148	0.9212	0.9247	0.9266	0.9315	0.9347	0.9382	9	0.801										
8	0.8168	0.886	0.9046	0.9142	0.9213	0.9286	0.9325	0.9357	0.9395			8	0.8166	0.8795	0.9001	0.9101	0.917	0.9222	0.9271	0.9282	0.9336	0.9367	0.94	8	0.8038										
7	0.8163	0.8855	0.905	0.9138	0.9217	0.9291	0.9331	0.9363	0.9402			7	0.8149	0.8808	0.9011	0.9114	0.9178	0.9243	0.9283	0.9303	0.9357	0.939	0.9408	7	0.8073	0.8754	0.8952	0.9057	0.9131	0.9189					
6	0.8158	0.8845	0.9052	0.9144	0.9226	0.9295	0.9331	0.9367	0.9405	0.944	0.9456	6	0.8148	0.8825	0.9017	0.912	0.9189	0.9259	0.9299	0.9325	0.9366	0.9398	0.9418	6	0.812	0.879	0.8984	0.9081	0.9155	0.9217					
5	0.8149	0.8842	0.9051	0.915	0.923	0.9291	0.9339	0.9375	0.9424	0.9448	0.9463	5	0.814	0.8827	0.9026	0.9125	0.9199	0.9269	0.9314	0.9335	0.9344	0.9411	0.9434	5	0.8131	0.88	0.8996	0.9112	0.9169	0.9241	0.9281	0.9307			
4		0.9047	0.9156	0.9228	0.929	0.9339	0.9382	0.9426	0.9454	0.9462		4	0.8122	0.8823	0.903	0.913	0.9211	0.9278	0.9319	0.9355	0.9394	0.9431	0.9449	4	0.8115	0.8817	0.9013	0.9117	0.9193	0.9261	0.9308	0.9322	0.9369	0.9411	0.9425
3		0.9033	0.9146	0.922	0.9291	0.9346	0.9389	0.943	0.9458	0.9462		3	0.8103	0.8822	0.9026	0.9139	0.9211	0.9278	0.9331	0.9368	0.9412	0.9441	0.9454	3	0.8101	0.8812	0.9017	0.9124	0.9207	0.9271	0.9317	0.9349	0.9391	0.9427	0.9444
2		0.9012	0.914	0.9205	0.9287	0.9347	0.9383	0.9432	0.9454	0.9466		2	0.8043	0.878	0.9001	0.9132	0.9194	0.9282	0.9334	0.9378	0.942	0.9445	0.9458	2	0.8064	0.8788	0.9	0.9124	0.9192	0.9277	0.9325	0.9362	0.9411	0.9439	0.9449
1			0.9088	0.9174	0.9279	0.9332	0.9372	0.9431	0.9455	0.9464		1	0.787	0.8671	0.895	0.9094	0.9175	0.9275	0.9328	0.9367	0.9429	0.945	0.9459	1	0.7883	0.8685	0.8954	0.9094	0.9175	0.9272	0.9324	0.9359	0.9424	0.9441	0.9454

MSE

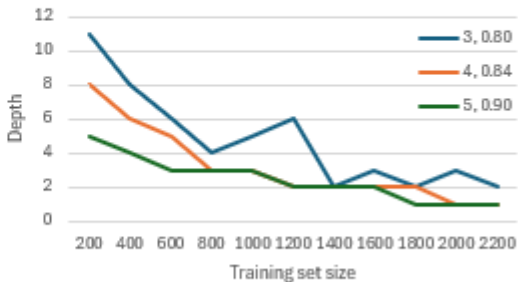
σ^2 of w and b = 3.0 and 0.8												σ^2 of w and b = 4.0 and 0.84												σ^2 of w and b = 5.0 and 0.9											
Training set size												Training set size												Training set size											
depth	200	400	600	800	1000	1200	1400	1600	1800	2000	2200	depth	200	400	600	800	1000	1200	1400	1600	1800	2000	2200	depth	200	400	600	800	1000	1200	1400	1600	1800	2000	2200
13	0.04029											13												13											
12	0.04003	0.03104	0.02699	0.02472								12												12											
11	0.03976	0.03079	0.02677	0.02451								11												11											
10	0.0395	0.03054	0.02654	0.0243	0.0225	0.021	0.02005					10	0.04352	0.0343	0.03001	0.02751	0.02559	0.02397	0.02291	0.02208	0.0212	0.02042	0.01984	10											
9	0.03925	0.0303	0.02631	0.02409	0.02229	0.0208	0.01985					9	0.04247	0.03339	0.02919	0.02676	0.02487	0.02328	0.02225	0.02144	0.02058	0.01983	0.01926	9	0.04591										
8	0.03903	0.03007	0.02609	0.02388	0.02209	0.02059	0.01965	0.01891	0.01816			8	0.04152	0.03254	0.02842	0.02606	0.02419	0.02263	0.02163	0.02084	0.01999	0.01926	0.01872	8	0.04421										
7	0.03885	0.02987	0.02589	0.02369	0.02189	0.02039	0.01945	0.01872	0.01798			7	0.04069	0.03178	0.02773	0.02542	0.02357	0.02203	0.02104	0.02026	0.01945	0.01873	0.01821	7	0.04271	0.03372	0.02955	0.02713	0.02522	0.02362					
6	0.03874	0.02971	0.02571	0.02351	0.0217	0.0202	0.01927	0.01853	0.01781	0.01715	0.01668	6	0.04001	0.03112	0.02711	0.02484	0.023	0.02147	0.0205	0.01974	0.01894	0.01825	0.01773	6	0.04145	0.03258	0.0285	0.02616	0.02428	0.02271					
5	0.03876	0.02963	0.02559	0.02338	0.02155	0.02003	0.0191	0.01837	0.01765	0.017	0.01653	5	0.03952	0.0306	0.0266	0.02435	0.02251	0.02098	0.02002	0.01926	0.01849	0.01781	0.01731	5	0.04047	0.03163	0.02762	0.02533	0.02346	0.0219	0.02092	0.02015			
4		0.02557	0.02331	0.02145	0.01991	0.01898	0.01824	0.01754	0.01689	0.01642		4	0.03932	0.03028	0.02623	0.02398	0.02211	0.02057	0.01962	0.01887	0.01812	0.01745	0.01696	4	0.03985	0.03095	0.02693	0.02466	0.02278	0.02122	0.02025	0.01949	0.01871	0.01801	0.01751
3		0.02573	0.02338	0.02146	0.01988	0.01894	0.01818	0.01749	0.01683	0.01635		3	0.03956	0.03028	0.0261	0.02378	0.02186	0.02029	0.01934	0.01858	0.01785	0.01718	0.01669	3	0.03975	0.03063	0.02652	0.02421	0.02228	0.02071	0.01974	0.01898	0.01822	0.01754	0.01704
2		0.02629	0.02373	0.02169	0.02003	0.01906	0.01828	0.01757	0.01689	0.01639		2	0.04066	0.03091	0.0264	0.0239	0.02188	0.02023	0.01925	0.01847	0.01774	0.01706	0.01657	2	0.04054	0.03098	0.02657	0.02411	0.02209	0.02045	0.01947	0.01869	0.01794	0.01725	0.01675
1			0.02487	0.02254	0.02067	0.0196	0.01874	0.01798	0.01723	0.01667		1	0.04408	0.03331	0.02786	0.02483	0.02254	0.02069	0.01963	0.01878	0.01801	0.01727	0.01672	1	0.04367	0.03308	0.02779	0.02485	0.02257	0.02075	0.01969	0.01884	0.01806	0.01733	0.01678

Plot

< Depth where maximum accuracy occurs for each training set size>

σ^2 of w and b			
Training set size	3, 0.80	4, 0.84	5, 0.90
200	11	8	5
400	8	6	4
600	6	5	3
800	4	3	3
1000	5	3	3
1200	6	2	2
1400	3	2	2
1600	2	2	2
1800	2	2	1
2000	3	1	1
2200	2	1	1

Change of depth for the best model by Accuracy with given training set size



< Depth where minimum MSE occurs for each training set size>

σ^2 of w and b			
Training set size	3, 0.80	4, 0.84	5, 0.90
200	9	4	3
400	5	3	3
600	4	3	3
800	4	3	2
1000	4	3	2
1200	3	2	2
1400	3	2	2
1600	3	2	2
1800	3	2	2
2000	3	2	1
2200	3	2	1

Change of depth for the best model by MSE with given training set size

